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Analysis and research on library user behavior based on apriori algorithm

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ABSTRACT

This paper discusses the analysis and research of library user behavior based on Apriori optimization algorithm. This algorithm can be used to analyze the borrowing behavior of users in the library. This article introduces the principle and process of the Apriori algorithm, and then performs data preprocessing and cleaning on the library borrowing records. Frequent itemsets and association rules are mined by using Apriori algorithm, and the results are explained and analyzed. The research results of this paper show that there is a certain correlation between the books borrowed by library users. According to the mining results of association rules, the library can recommend different books for different types of users, so as to improve user satisfaction and borrowing rate.

1. Introduction

Big data technology has become an integral part of today's society, and libraries are no exception. With the increasing number of digital resources, big data technology is helping libraries better organize and utilize information and improve service quality [1]. First, big data technology can help libraries better understand readers' needs and preferences. By analyzing readers' borrowing records, search history, and online reviews, libraries can better understand readers' reading preferences and provide them with better services. Secondly, big data technology can also help libraries improve resource utilization efficiency [2]. For example, by analyzing a library's borrowing data, a library can determine which titles are most popular and whether to increase its holdings. In addition, big data technology can also help libraries better manage library collections. For example, by analyzing the use of library collections, libraries can decide whether to repair or update books [3].

Association rule mining is the process of mining valuable knowledge that describes the relationships between data items from a large amount of data. As one of the most classic methods in data mining, it is widely used in various industries [4–6]. For example, when applied to the commercial retail industry, analyze the purchasing characteristics of customers to guide the shelf arrangement, pricing, and purchase of commodities; when applied to the financial and insurance service industries, by analyzing a large amount of financial data, establish trade and risk models for investment activities, Guiding stock selection, currency trading, etc.; applied to the field of scientific research, useful information can be extracted from a large amount of scattershot scientific data; also used in telecommunications network management, medical data mining, manufacturing fault diagnosis, etc. application field [7–9]. At present, university libraries generate a large amount of book circulation data every day, and the use of association rule analysis can also dig out useful information and implicit information from a large number of historical borrowing data of readers to guide library services and management [10,11].

The Apriori algorithm is the first association rule mining algorithm and the most classic algorithm. It uses an iterative method of layer by layer search to find the relationships between item sets in the database, forming rules. The process consists of concatenation (class matrix operation) and pruning (removing unnecessary intermediate results). The concept of item sets in this algorithm is the set of items.

This paper uses the borrowing data generated by the library information system to conduct in-depth mining through the Apriori algorithm and the K-means algorithm, so as to discover this "hidden correlation", so as to promote the service work of the university library towards the direction of intelligence. A large number of experiments were conducted to verify the effectiveness of the improved Apriori algorithm.

The rest of this paper is organized as follows: Section 2 analyzed the library information management system. Through the elaboration of the overall architecture, the entire process of data processing was further analyzed, including data cleaning, data integration, data transformation, data reduction, and other operations. Section 3 shows the experimental analysis, and Section 4 concludes the paper with summary

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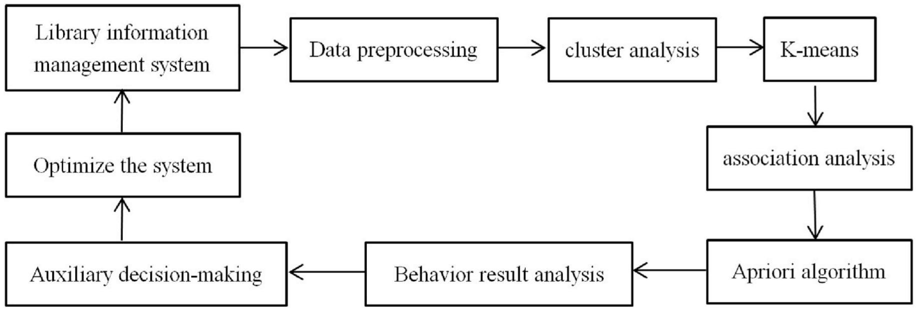

Fig. 1. Overall architecture.

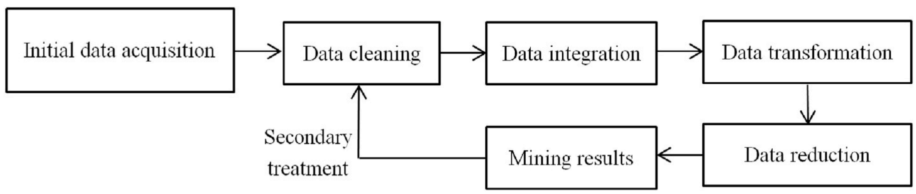

Fig. 2. Data preprocessing.

and future research directions.

2. Library information management model

Traditional library user data analysis is based on data statistics, and realizes functions such as data retrieval and classification. A large amount of borrowing data in the library database can well reflect the needs of users through data mining, and find out the association rules between books borrowed by users, the association rules between books of different disciplines, and the laws of books borrowed by users of different specialties etc [12–14].

The overall architecture of the library information management system is shown in Fig. 1. Extract user information such as students and teachers from the library information system and establish a user behavior database [15]. Establish a data standard system for dimensions such as user borrowing records, interests, and habits through preprocessing processes such as data cleaning, transformation, and intensive processing [16]. Classify user tags through clustering analysis and K-means algorithm, use Apriori association rules to analyze user behavior association rules and patterns, identify and analyze user characteristics and their borrowing behavior patterns, and finally discuss users' hidden needs, behavioral habits, and future borrowing trends. Ultimately, achieve the goal of personalized information services in a true sense, and serve university library users [17–19].

2.1. Data preprocessing

Data processing includes data cleaning, data integration, data transformation, and data reduction of the initial data. If the fused data does not meet the requirements, a second round of data cleaning and other operations will be carried out. Data preprocessing is as shown in Fig. 2.

Data cleaning is an important part of data preprocessing and one of the key steps in data analysis. The main purpose of data cleaning is to process data so that it can be accurately analyzed and used in order to better extract useful information. Data cleaning includes the steps of deleting duplicate data, missing value processing, outlier value processing, data type conversion, data format unification, and data normalization [20].

Data integration is the process of merging data from multiple sources into one dataset. Data integration includes the steps of identifying data sources, identifying and resolving schema conflicts, identifying and resolving data conflicts, data transformation, data quality checks, and data integration management [21].

Data transformation refers to the processing and transformation of raw data to support tasks such as data analysis, data mining, and machine learning. Data transformation includes steps such as data cleaning, data formatting, data normalization, data aggregation, feature extraction, and data dimensionality reduction [22].

Data reduction refers to reducing the amount of data by removing redundant information and retaining important information, thereby reducing the cost of data processing and storage, while retaining enough information to support tasks such as data analysis and mining. Data reduction includes the steps of data sampling, feature selection, feature extraction, data compression, and data reduction.

2.2. K-means cluster analysis

Cluster analysis is a data mining technique that divides objects in a data set (for example, text, images, or numerical data) into groups such that the similarity between objects in each group is maximized, and minimize the similarity between them. Cluster analysis is often used to explore patterns and structures in data sets and to help determine how to classify or group data, which can aid in decision-making in a variety of application domains.

K-means is a classic clustering algorithm, which divides the data set into K different clusters, and each cluster represents a group of similar data objects. The core idea of K-means is to divide the data into K clusters, so that the similarity between data objects in the same cluster is as high as possible, while the similarity between different clusters is as low as possible. In this paper, Ransomian distance is used for distance measurement.

Rankine distance is a distance measurement method used to compare the difference between two sequences, and it is used to compare whether the sorting relationship between elements in the two sequences is consistent. Rankine distance is often used to evaluate the performance of classification models, or to measure the similarity between data points in cluster analysis.

The calculation method of Ran's distance is: first sort the elements in the two sequences according to their size, then compare their relative positions, and calculate the reverse logarithm between them (that is, the elements in one sequence are arranged in the other sequence The number of times before the corresponding element), and finally divide the logarithm of the reverse order by the total number of elements in the sequence to obtain the Langerian distance. The value range of Ran's distance is from 0 to 1, where 0 means that the two sequences are completely consistent and 1 means that the two sequences are completely different.

When $x_{ij} > 0(i = 1, 2, \cdots, n; j = 1, 2, \cdots, p)$, define the Ransomian distance between the i sample x_i and the j sample x_j as:

$$d_{ik}(L) = \sum_{k=1}^{p} \frac{|x_{ik} - x_{jk}|}{x_{ij} + x_{jk}}, i = 1, 2, \cdots, n; j = 1, 2, \cdots, n \tag{1}$$

The advantage of the Rankine distance is that it is not sensitive to the size and scale of the data, so it is suitable for processing data of different scales. In addition, it can be used to evaluate the ranking performance of classification models, not just accuracy, thus more comprehensively evaluating the performance of models.

The similarity coefficient is measured by the correlation coefficient.

$$C_{ij}(2) = \frac{\sum_{k=1}^{n}(x_{ik} - \overline{x_i})(x_{jk} - \overline{x_j})}{\sqrt{\left[\sum_{k=1}^{n}(x_{ik} - \overline{x_i})^2\right]\left[\sum_{k=1}^{n}(x_{jk} - \overline{x_j})^2\right]}}, i = 1, 2, \cdots, n; j = 1, 2, \cdots, n \tag{2}$$

In which, $\overline{x_i} = \frac{1}{n}\sum_{k=1}^{n} x_{ik}, \overline{x_j} = \frac{1}{n}\sum_{k=1}^{n} x_{jk}, i = 1, 2, \cdots, n; j = 1, 2, \cdots, p.$

The steps of the K-means algorithm include:

Step 1: Randomly select K points as the initial cluster centers;
Step2: For each data object, calculate its distance to the center of each cluster and divide it into the nearest cluster;
Step3: For each cluster, recalculate its cluster center;
Step4: Repeat step 2 and step 3 until the cluster center does not change or reaches the predetermined number of iterations. This iteration minimizes the following formula.

$$\mathrm{E} = \sum_{j=1}^{k} \sum_{x_i \in \omega_j} \|x_i - m_j\|^2 \tag{3}$$

2.3. Apriori optimization algorithm

Association rule learning is a data mining technique aimed at discovering implicit association rules from a large amount of transactional data. These rules can be expressed in the form of X→Y, which means that when itemset X appears, itemset Y is also likely to appear. The purpose of association rule learning is to find frequent itemsets and mine association rules from them. The application of association rule learning in the library field includes reader behavior analysis, book recommendation system, etc. For example, readers' interest association rules can be mined by analyzing readers' historical borrowing records, so as to better recommend related books to readers. Common association rule learning algorithms include Apriori algorithm and Eclat algorithm.

The Apriori algorithm is an algorithm proposed in 1994 for mining association rules between data. In the field of libraries, the Apriori algorithm can be used to analyze readers' reading behavior. By analyzing readers' historical borrowing records, the Apriori algorithm can find books that readers often borrow together. These association rules can be used as patterns of reader behavior to recommend relevant books to readers. For example, assuming that a reader often borrows books of history, culture, and biography, then an association rule can be obtained by analyzing the reader's borrowing behavior through the Apriori algorithm: if the reader borrows books of history, then he is very It is possible to borrow books on culture and biography. Therefore, through the Apriori algorithm, libraries can better understand readers' reading behavior and provide readers with more personalized services.

The Apriori algorithm is a recursive algorithm based on frequent set search and frequent set rule generation process, the goal is to find the largest frequent set of K items. Whether the frequent set is suitable is evaluated according to the support and confidence.

Table 1
List of database transactions.

transactions	ID list
T100	I1,I2,I5
T200	I2,I4
T300	I2,I3
T400	I1,I2,I4
T500	I1,I3
T600	I2,I3
T700	I1,I3
T800	I1,I2,I3,I5
T900	I1,I2,I3

The support degree of an association rule is the ratio of the number of transactions of X and Y included in the transaction set to the number of all transactions, which is shown as:

$$support(X \rightarrow Y) = support(X \cup Y) = P(XY) \tag{4}$$

The confidence of an association rule is the ratio of the number of transactions containing X and Y to the number of all transactions containing X in the transaction set, denoted as:

$$confidence(X \rightarrow Y) = \frac{support(X \cup Y)}{support\ (X)} = P(Y|X) \tag{5}$$

Support reflects the probability that the items contained in X and Y appear in the transaction set at the same time. Confidence reflects the conditional probability of occurrence of Y in a transaction containing X. That is, the proportion of support $(X \cup Y)$ to support (X).

The steps of the Apriori algorithm are:

Step 1: Scan the entire data warehouse to calculate all frequent itemsets, and find frequent itemsets consistent with the previously set support;
Step2: Based on the rules for generating frequent itemsets in the previous step, by scanning the support of candidate frequent K-itemsets, discard the data sets whose support in candidate frequent K-itemsets is lower than the threshold;
Step3: Get frequent K-itemsets suitable for the actual needs of the project. The result obtained through K iterations may be an empty set, and the frequent set of K-1 items is returned as the final result.

The following example illustrates the whole process of Apriori algorithm. Table 1 is a transaction list of a database, there are 9 sets of data in the database, $|D| = 9$. Set the minimum support count to 2, min_sup = 2.

First scan: Scan database D to obtain counts for each candidate. No items were deleted due to a minimum transaction support of 2. The frequent 1-item set L1 can be determined, which is composed of candidate 1-item sets with the minimum support, as shown in Fig. 3.

Second scan: In order to find the set L2 of frequent 2-itemsets, the algorithm uses the L1∞L1 algorithm to generate the set C2 of candidate 2-itemsets, and then obtain L2 according to the support, as shown in Fig. 4.

Third scan: L2∞L2 produces a set C3 of candidate 3-itemsets.

$$\mathrm{C3} = \mathrm{L2} \infty \mathrm{L2} = \{\{I1, I2\}, \{I1, I3\}, \{I1, I5\}, \{I2, I3\}, \{I2, I4\}, \{I2, I5\}\} \infty$$

$$\{\{I1, I2\}, \{I1, I3\}, \{I1, I5\}, \{I2, I3\}, \{I2, I4\}, \{I2, I5\}\}$$

$$= \{\{I1, I2, I3\}, \{I1, I2, I5\}, \{I1, I3, I5\}, \{I2, I3, I4\}, \{I2, I3, I5\}, \{I2, I4, I5\}\}$$

Because all non-empty elements of frequent itemsets must also be frequent sets, C3 $= \{\{I1, I2, I3\}, \{I1, I2, I5\}\}$, as shown in Fig. 5.

Fourth scan:L3∞L3 produces a set C4 of candidate 4-itemsets.

C1

Itemset	Support count
I1	6
I2	7
I3	6
I4	2
I5	2

$\xrightarrow[\text{minimum support count}]{\text{Compare candidate support count}}$

L1

Itemset	Support count
I1	6
I2	7
I3	6
I4	2
I5	2

Fig. 3. Support count for the first scan.

C1

Itemset
I1,I2
I1,I3
I1,I4
I1,I5
I2,I3
I2,I4
I2,I5
I3,I4
I3,I5
I4,I5

$\longrightarrow$

C2

Itemset	Support count
I1,I2	4
I1,I3	4
I1,I4	1
I1,I5	2
I2,I3	4
I2,I4	2
I2,I5	2
I3,I4	0
I3,I5	1
I4,I5	0

$\longrightarrow$

L2

Itemset	Support count
I1,I2	4
I1,I3	4
I1,I5	2
I2,I3	4
I2,I4	2
I2,I5	2

Fig. 4. Support count for the second scan.

C3

Itemset
I1,I2,I3
I1, I2,I5

$\longrightarrow$

C3

Itemset	Support count
I1,I2,I3	2
I1, I2,I5	2

$\longrightarrow$

L3

Itemset	Support count
I1,I2,I3	2
I1, I2,I5	2

Fig. 5. Support count for the third scan.

$L3 \infty L3 = \{\{I1, I2, I3, I5\}\}$

Since the subset $\{I2, I3, I5\}$ is not frequent, this itemset is deleted, and $C4 = \infty$ is obtained, and the algorithm terminates.

3. Experimental results and analysis

The data comes from some publicly available library data. It includes the user's basic information (fields such as student number, grade, major, subject course, etc.) and book borrowing information (fields such as title, call number, China Library classification number, author, publication, etc.), through which the user's borrowing behavior can be described.

Before optimizing the allocation of library collections, books are divided into n categories according to their categories, which are recorded as T1, T2, …, Tn, and the historical borrowing information of n categories of books is stored in the database TL for book management. At the same time, it is assumed that currently, there are k bookshelf areas in the library that needs to be optimized. Now we are looking for the best configuration of n types of books, so that readers can find the required number as conveniently as possible according to the expected demand, so that the borrowing distance of books in one borrowing process is the shortest.

Table 2
User borrowing transaction data table.

TID	A	B	C	D	E	F	G	⋯	Z
001	0	0	1	4	0	2	1	⋯	0
002	0	0	2	2	1	1	2	⋯	1
003	1	1	0	2	3	1	0	⋯	0
004	2	3	4	0	2	0	2	⋯	0
005	1	1	4	1	2	1	1	⋯	2
006	0	1	2	2	3	0	1	⋯	2
⋮									⋮
007	2	0	2	1	1	1	4	⋯	2
008	2	1	1	0	3	1	0	⋯	1
009	1	2	3	0	0	0	2	⋯	1
010	0	3	2	0	1	3	4	⋯	0

The collected data is cleaned by data preprocessing. Classify the borrowed books according to the Chinese library method, so as to integrate and merge the user's borrowed books information. The processing results are shown in Table 2 (some representative data).

Using the user borrowing transaction data table, the K-means clustering algorithm is used to segment users to generate user similarity groups. This paper clusters the number of times that users borrow

Table 3
K-means results.

Classification	Class 1	Class 2	Class 3	Class 4	Class 5
variance	7.32	10.1	13.1	18.7	47.6
Borrowed quantity	4.6	12.5	18.1	38.1	52.3
Number of users	5212	3832	1611	71	7

various books, and divides users into clusters with common behaviors. The types and quantities of books borrowed by users are relatively similar among the clusters, which are reflected in the difference in the number of books borrowed by users and the frequency of borrowing by users. In the actual K-means clustering process, through continuous adjustment of the K value, the K value is finally determined to be 5, that is, users are subdivided into 5 categories. The results of user group cluster analysis are shown in Table 3, and the proportion of the number of books used is shown in Fig. 6. From the table, it can be seen that class 1 has the highest number of users, the lowest variance, and the lowest borrowed quantity. Class5 has the lowest number of users, the highest variance, and the highest borrowed quantity.

In this cluster analysis, the total number of user groups is 10,733 users. From the clustering results in the figure, it can be seen that category 1 has the most users, accounting for 48.6%, with 5212 users, and the number of books borrowed in this year is 4.6; the number of users in cluster 2 is 3832, and the number of users in cluster 3 The number of users is 1611, the number of users in cluster 4 is 71, and the number of users in cluster 5 is 7. It can be seen that cluster 1, cluster 2, cluster 3, and cluster 4 include the vast majority of users, accounting for 99.9%, showing that the vast majority of users have an annual borrowing list of 4.6–38.1 books, which also shows that the school's overall books Utilization is low. The number of users in cluster 5 is 7, which is an extremely rare case and not universal.

Apriori algorithm is an algorithm commonly used in association rule mining, but it also has the following disadvantages:

Generation of a large number of candidate item sets: Apriori algorithm needs to generate all possible candidate item sets, including those that do not meet the minimum support, which will result in a large amount of computing and storage overhead;

The number of frequent itemset scans is large: the Apriori algorithm needs to scan the data set multiple times to determine frequent itemsets, and these scan operations may take up a lot of time and computing resources;

The performance of the algorithm is limited by the density of the data: when the data set is very dense, the Apriori algorithm will generate a large number of candidate item sets, and most of these candidate item sets are not frequent item sets, resulting in low algorithm efficiency;

Not suitable for processing large-scale data sets: When the size of the data set exceeds the memory capacity, the performance of the Apriori algorithm will be severely limited, or even unable to complete the task.

In order to solve this problem, this paper makes the following optimizations:

In the Apriori algorithm, frequent itemsets must meet the requirements of support, and infrequent itemsets cannot become frequent itemsets. Therefore, infrequent item sets can be deleted through pruning operations to reduce the amount of calculation and storage space;

Apriori algorithm uses candidate itemsets to generate frequent itemsets, and the size of candidate itemsets depends on the number of items in the data set. In order to reduce the number of candidate item sets, set compression technology can be used to combine multiple items into one item to reduce the number of candidate item sets;

Apriori algorithm needs to scan the data set multiple times when searching for frequent itemsets. In order to reduce the number of scans, a technique based on a hash table can be used to save frequent item sets in a hash table and use the hash table to find frequent item sets to improve efficiency.

In this paper, the minimum support degree min_sup = 0.02 is set. In multiple experiments, min_sup = 0.03 was finally selected, and some of the results obtained are shown in Table 4.

It can be seen from the table that 3.35% of the readers borrowed TN and TM books while borrowing TP series books. Among the readers who have borrowed TP, 77.54% have borrowed TN and TM books. 4.42% of the readers borrowed TP and TU books while borrowing TN series books. Among the readers who have borrowed TN, 77.54% have borrowed TP and TU books.

The improved April algorithm has faster execution time compared to the traditional April algorithm, as shown in Fig. 7. From the figure, it can be seen that when using the same support and confidence levels, as the number of transactions continues to increase, the execution speed of the improved algorithm significantly accelerates. The improved algorithm has no increase in computational complexity and can be executed more than 300 times per second.

4. Conclusions

A reasonable layout and configuration of library collections is beneficial for readers to efficiently achieve book borrowing and provide

Table 4
Association result.

consequent	Previous	Support	confidence
TP	TN TM	0.0335	77.54
TN	TP TU	0.0442	78.44

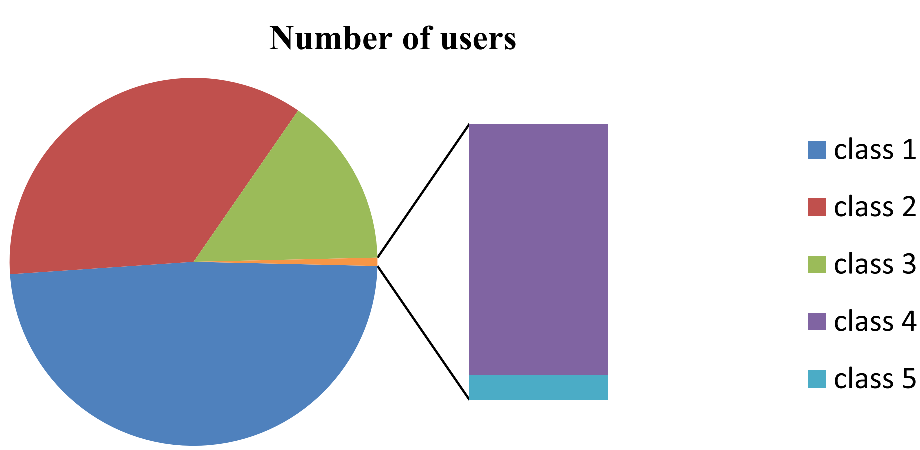

Fig. 6. Number of users scale chart.

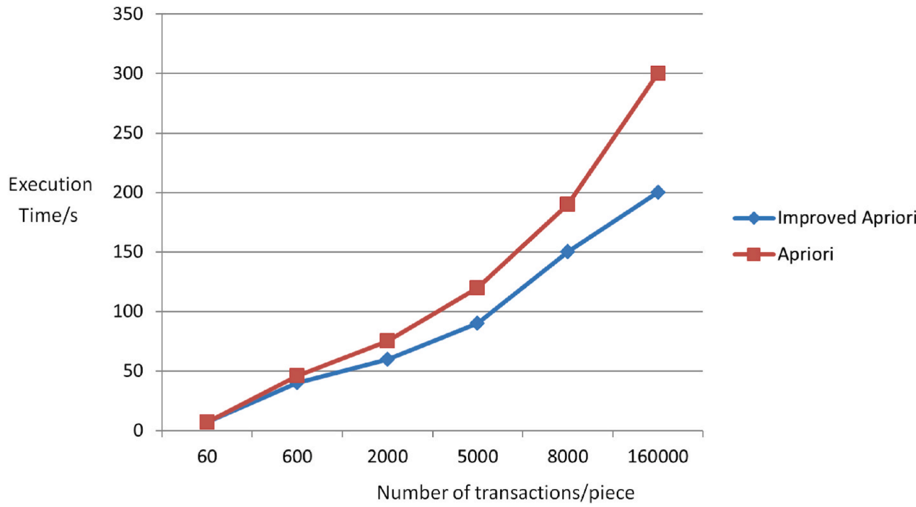

Fig. 7. Comparison chart.

them with good book borrowing expectations. This article conducts research on the layout of library collections and user borrowing habits, mining effective book association relationships from historical borrowing data. Transforming the collection layout problem into a confidence calculation problem for simultaneously borrowing books. A library collection configuration optimization algorithm based on the combination of April algorithm and k-means clustering algorithm was designed, and the effectiveness of the method was verified. It has important application value for library user behavior analysis and research, and can provide useful reference and guidance for libraries. Further optimization of the algorithm will be carried out in the future to achieve higher efficiency.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

Data availability

Data will be made available on request.

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